

ITAI-1371-ML-Dataset: Exploratory Data Analysis & Real Estate Price Modeling

Team: 14274 2_ITAI1371 | Course: Mid Term - Exploratory Data Analysis

Introduction

This paper outlines the midterm project plan for ITAI-1371, focusing on Exploratory Data Analysis (EDA) and real estate price modeling using the "Connecticut Real Estate Sales 2001-2021" dataset from Kaggle. Our team—Marisabel Morales, Katherine Delores, and Joshua Balderas—has defined a workflow for data preparation, visualization, and future machine learning application.

Dataset and Feature Selection

The refined dataset contains 25,000 records and 10 selected features, hosted on GitHub. Following instructor consultation, we selected:

- Identifiers & Temporal: Serial Number, List Year, Date Recorded
- Geographic & Location: Town, Address
- Valuation & Financials: Assessed Value, Sale Amount, Sales Ratio
- Classification Attributes: Property Type, Residential Type

Metadata identifiers and trailing blank columns were programmatically excluded in Python.

Planned Pre-Processing Methodology

1. Train/Test Split: A 70% training / 30% testing split using Scikit-Learn; the test partition remains untouched in cold storage.
2. Missing Value Imputation: Missing entries in 'Property Type' and 'Residential Type' will be imputed with 'Unspecified' and 'Non-Residential'; null addresses filled with 'Unknown Address'.
3. Categorical Encoding: One-hot encoding for nominal categories ('Property Type', 'Residential Type').
4. Normalization & Scaling: Log-transformation ($\log_{1p}$) applied to skewed monetary fields ('Sale Amount', 'Assessed Value'), followed by StandardScaler standardization.
5. Visual Verification: Matplotlib side-by-side histograms comparing raw vs. normalized distributions and class balance counts.

Proposed Downstream Machine Learning Problems

- Supervised Regression: Train models (Linear Regression, Random Forest, XGBoost) to predict 'Sale Amount' using assessed valuations, location, and property characteristics.
- Binary Classification: Classify transactions as undervalued ('Sales Ratio' < 0.70) versus standard market valuation.

Team Roles

- Katherine Delores: Team Contribution Journal
- Joshua Balderas: Jupyter Notebook
- Marisabel Morales: Reflective Journal

This division ensures all project aspects—from technical implementation to reflective analysis—are thoroughly addressed.